Week 1 Concentration

Ref: Vershynin Chap 2&3

Concentration inequality of a random variable X is of the form:

$$\mathbb{P}[|X-\mu|>t] \leq \text{sth. small}$$

Often, in high-dimensional problems (eg. X = norm of high-dim random matrix) concentration can be established by systematic tools.

Example: S_N = # of heads in N coin tosses.

Chebyshev ineq: $\mathbb{P}\left[\left|\frac{S_N}{N}-\frac{1}{2}\right|>t\right] \leq \frac{\mathrm{Var}(S_N/N)}{t^2} = O\left(\frac{1}{Nt^2}\right)$

CLT suggests $\mathbb{P}\left[2\sqrt{N}\left|\frac{S_N}{N}-\frac{1}{2}\right|>2\sqrt{N}t\right] = O\left(e^{-4Nt^2}\right)$

not correct for $t = \Theta(1)$!

However we can use nonasymtotic concentration inequalities (Hoeffding, Chernoff, Bernstein, McDiarmid, log-Sobolev transportation-cost) to show an upper bound of the form:

$$O(e^{-cNt^2})$$

for $c>0$.

- Sub-Gaussian Let X be random r.v. the following definitions are equivalent:

(i) $\exists\ K_1>0$ such that

$$P[|X|\geq t] \leq 2\exp(-t^2/K_1^2) \quad \forall\ t\geq 0$$

(ii) $\exists\ K_2 > 0$ s.t.

$$\|X\|_{L_p} \le K_2 \sqrt{p} \quad \forall\ p \ge 1$$

(iii) $\exists\ K_3 > 0$ s.t.

$$\mathbb{E} \exp(\lambda^2 X^2) \le \exp(K_3^2 \lambda^2) \quad \forall\ \lambda \in \left[-\frac{1}{K_3}, \frac{1}{K_3}\right]$$

(iv) $\exists\ K_4 > 0$ s.t.

$$\mathbb{E}\left[\exp\left(X^2 / K_4^2\right)\right] \le 2$$

(v) (If $\mathbb{E}X = 0$) $\exists\ K_5 > 0$ s.t.

$$\mathbb{E}\left(\exp \lambda (X - \mathbb{E}X)\right) \le \exp\left(K_5^2 \lambda^2\right), \quad \forall\ \lambda \in \mathbb{R}$$

Defn: sub-Gaussian norm:

$$\|X\|_{\psi_2} = \inf\left\{ t>0 ;\ \mathbb{E}\exp\left(\frac{X^2}{t^2}\right) \le 2 \right\}$$

Exercise: if $X_1, X_2, \cdots$ are sub-Gaussian (possibly dependent), $K = \max_i \|X_i\|_{\psi_2}$. Then

$$\mathbb{E}\left[\max_{i\le N} |X_i|\right] \le CK\sqrt{\log N}$$

- sub-exponential: square of sub-Gaussian
  natural in large deviation analysis

(a) $\mathbb{P}[|X| \ge t] \le 2\exp(-t/K_1)$ $\quad \forall t \ge 0$

(b) $\|X\|_{L^p} = (\mathbb{E}|X|^p)^{\frac{1}{p}} \le K_2 p$, $\quad \forall p \ge 1$

(c) $\mathbb{E}\exp(\lambda|X|) \le \exp(K_3\lambda)$, $\forall \lambda \in [0, K_3]$

(d) $\mathbb{E}\exp(|X|/K_4) \le 2$

(e) (If $\mathbb{E}X = 0$)

$$\mathbb{E}\exp\left(\lambda(X - \mathbb{E}X)\right) \le \exp\left(K_5^2 \lambda^2\right), \quad \forall\, \lambda \in \left[-\frac{1}{K_5}, \frac{1}{K_5}\right]$$

Defn: sub-exponential norm:

$$\|X\|_{\psi_1} = \inf\left\{t > 0 ;\ \mathbb{E}\exp(|X|/t) \le 2\right\}.$$

Lem: $\|X^2\|_{\psi_1} = \|X\|_{\psi_2}^2$

Exercise: suppose $X_1, \cdots, X_n \sim P$ are iid. under which assumptions do we have

$$\mathbb{P}\left(\left|\frac{X_1 + \cdots + X_n}{N} - \mathbb{E}X_1\right| > c^{-}\right] = o(e^{-cn})$$

for some $c > 0$ independent of n?

① $\|X_1\|_{\psi_1} < \infty$

② $\|X_2\|_{\psi_2} < \infty$

③ $\mathrm{Var}(X_1) < \infty$

④ $\|X_1\|_p < \infty$

answer: ① and ② see Bernstein inequality and Chernoff bound.

Bernstein inequality: Let $X_1, \cdots, X_N$ be independent, mean-zero. $\max_i \|X_i\|_{\psi_1} \le 1$. Then for every $t \ge 0$,

$$\mathbb{P}\left[\left|\frac{1}{N}\sum_{i=1}^{N} X_i\right| \ge t\right] \le 2\exp\left[-c\min\left(t^2, t\right)N\right].$$

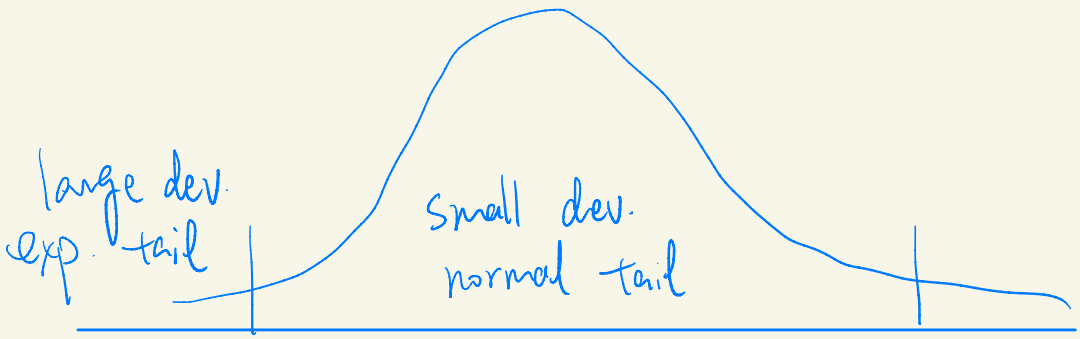
Remark: general vesions in Vershynin:

- $\frac{1}{N}\sum_{i=1}^{N} X_i \longrightarrow \sum a_i X_i$
- better bound when $\|X_i\|_{\psi_1}, i = 1, \cdots, N$ differ
- better bound using $\mathrm{Var}(X_i)$ when X_i finite

proof based on Chernoff's bound.

Example: let $X_i + 1 \sim \exp(1)$ i.i.d.

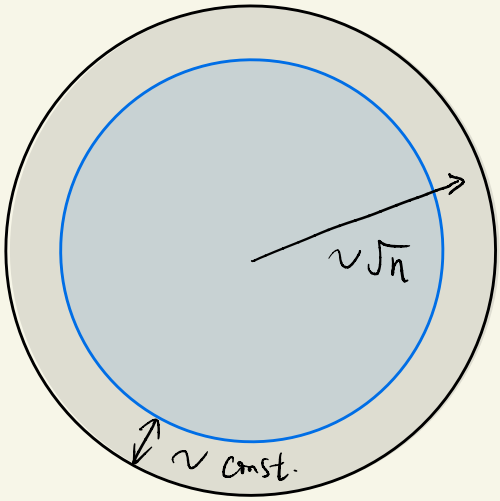
is $\mathbb{P}\left[\left|\frac{1}{N}\sum_{i=1}^{N} X_i\right| \geq t\right]$ more like Gaussian tail or exponential tail?



Thm 3.1.1 (concentration of the norm) $X = (X_1 \cdots X_n)$, independent coordinates. $K = \max_i \|X_i\|_{\psi_2} < \infty$, $\mathbb{E}X_i^2 = 1$. Then

$$\left\| \|X\|_2 - \sqrt{n} \right\|_{\psi_2} \leq CK^2$$

pf using Bernstein.



· sub-Gaussian random vector:

$$\|X\|_{\psi_2} := \sup_{x \in S^{n-1}} \|\langle X, x \rangle\|_{\chi_2}$$

Lem 3.4.2 X_i independent, $\mathbb{E}[X_i] = 0$, $X = (X_1 \cdots X_n)$

$$\Rightarrow \|X\|_{\psi_2} \le C \max_{i \le n} \|X_i\|_{\chi_2}$$

If independence is dropped, $\|X\|_{\psi_2} \le C\sqrt{n} \max_{i \le n} \|X_i\|_{\chi_2}$

- Example: uniform on $S^{n-1}(\sqrt{n})$

$$\Rightarrow \quad \|X\|_{\psi_2} = O(1)$$

Pf: Let $Y = (Y_1 \cdots Y_n)$ iid $\mathcal{N}(0,1)$

$$\mathbb{P}[Y_1 > \lambda] \geq \mathbb{P}\left[Y_1 > \lambda, \ \|Y\|_2 \geq \frac{\sqrt{n}}{2}\right]$$

$$\geq \mathbb{P}\left[\|Y\|_2 \geq \frac{\sqrt{n}}{2}\right] \inf_{\ell \geq \frac{\sqrt{n}}{2}} \mathbb{P}\left(Y_1 > \lambda \,\middle|\, \|Y\|_2 = \ell\right)$$

$$= \left(1 - o(1)\right) \cdot \mathbb{P}\left[Y_1 > \lambda \,\middle|\, \|Y\| = \frac{\sqrt{n}}{2}\right]$$

$$= \left(1 - o(1)\right) \mathbb{P}[X_1 \geq \lambda]$$

$$\Rightarrow \quad \mathbb{P}[X_1 > t) \leq \left(1 + o(1)\right) \mathbb{P}\left[Y_1 > \frac{t}{2}\right]$$

$$\leq \left(1 + o(1)\right) e^{-t^2/4}, \quad \forall t > 0$$

Remark 3.48 projective limit $x \in S^{n-1}$

$$\Rightarrow \quad \langle X, x \rangle \to \mathcal{N}(0,1) \text{ in distribution}$$

Example 3.44. uniform on $\{\sqrt{n}\, e_i\}_{i=1}^n$

$$\|X\|_{\psi_2} = O\left(\sqrt{\frac{n}{\log n}}\right)$$

Pf: let $|\lambda| \le \frac{1}{K}$ and $v = (v_1 \cdots v_n)$, $\|v\| = 1$

$$\Rightarrow \mathbb{E} \exp\left(\lambda^2 (\textstyle\sum X_i v_i)^2\right)$$

$$= \frac{1}{n} \sum_{i=1}^n \exp\left(\lambda^2 n v_i^2\right)$$

convexity

$$\le \frac{1}{n} \exp\left(\lambda^2 n\right) + \frac{n-1}{n}$$

$$= \exp\left(\lambda^2 n - \log n\right) + \frac{n-1}{n} \qquad (*)$$

Want: $|\lambda| \le \frac{1}{K} \Rightarrow \lambda^2 n - \log n \le \text{const}$

$$\Leftrightarrow \quad \frac{n}{K^2} - \log n \le \text{const}$$

$\Rightarrow$ can choose $K = \sqrt{\frac{n}{\log n}}$

Then $(*) \le 1 + \frac{n-1}{n} = 2$, for all $|\lambda| \le \frac{\sqrt{n}}{\sqrt{\log n}}$

Example: X uniform on $\mathrm{conv}\{ n e_i \}_{i=1}^{n}$

$\Rightarrow$ isotropic: $\mathbb{E}[X X^T] \approx 1$

$\|X\|_{\psi_1} = O(1)$

See Sec 3.4.4

$\mathbb{P}[X_1 > \frac{n}{2}] = e^{-O(n)}$